Utility Theory and the Loss Function

Stats 220B

Recap

In Lectures 1–2 we built the decision-theoretic framework: Θ, $\mathcal{A}$, $L(\theta, a)$, δ, $R(\theta, \delta)$, $\rho(\pi, a)$, $r(\pi, \delta)$, admissibility, three selection principles, randomized rules.

We proved three structural results: sufficiency reduces the problem (Theorem 1), convex loss eliminates randomization (Theorem 3), and the Rao–Blackwell theorem combines them.

Throughout, we used the loss function $L(\theta, a)$ without asking two questions:

- Where does the loss function come from?
- Why is *expected* loss the right way to aggregate it?

Today we address both. **Reading:** Berger, Chapter 2 (except Section 2.3).

Two problems with "value"

In evaluating consequences of actions, two problems arise:

1. Some consequences have no natural scale. Prestige, customer goodwill, reputation – important to businesses, but hard to quantify. A company considering selling its name-brand product in discount stores can estimate the immediate profit, but the long-term effect on prestige is much harder to measure.

2. Even when a monetary scale exists, it may not reflect true value. Suppose you can do an unpleasant task for $100. At your current income, you might do it. After receiving a million dollars, you probably would not. The value of $100 depends on context.

Value is not linear in money

Suppose you are offered a choice:

(A) Receive \$200,000 for certain.
(B) A 50–50 chance of receiving \$0 or \$410,000.

The expected monetary value of (B) is \$205,000 $>$ \$200,000. Yet most people prefer (A).

This means the "value" of money is not linear in the amount. The marginal value of money is decreasing: the difference in value between \$0 and \$200,000 is larger than the difference between \$200,000 and \$410,000.

We need a way to assign numerical values to consequences that reflects the decision maker's true preferences. That is what utility theory does.

The setup

Let $\mathcal{R}$ be the set of all possible consequences (rewards). Let $\mathcal{P}$ be the set of all probability distributions on $\mathcal{R}$.

We assume the decision maker can state preferences among elements of $\mathcal{P}$:

- $P_1 \prec P_2$: P_2 is preferred to P_1.
- $P_1 \approx P_2$: P_1 and P_2 are equivalent.
- $P_1 \preceq P_2$: P_1 is not preferred to P_2.

A reward $r \in \mathcal{R}$ is identified with the degenerate distribution (r) that puts probability one on r. So preferences apply to rewards as well.

The goal: find a function $U : \mathcal{R} \to \mathbb{R}$ such that
$P_1 \prec P_2 \iff E^{P_1}[U(r)] < E^{P_2}[U(r)]$. Such a U is called a **utility function**.

Constructing a utility function

Assume a utility function exists. Here is one way to build it.

Step 1. Choose two rewards $r_1 \prec r_2$. Set $U(r_1) = 0$, $U(r_2) = 1$. (This fixes the scale.)

Step 2. For any r_3 with $r_1 \prec r_3 \prec r_2$: find $\alpha \in (0, 1)$ such that $r_3 \approx \alpha(r_1) + (1 - \alpha)(r_2)$. Then $U(r_3) = 1 - \alpha$.
The idea: r_3 is equivalent to a gamble between r_1 (prob. α) and r_2 (prob. $1 - \alpha$). Then $U(r_3) = \alpha U(r_1) + (1 - \alpha)U(r_2) = 1 - \alpha$.

Step 3. For $r_3 \prec r_1$: find α such that $r_1 \approx \alpha(r_3) + (1 - \alpha)(r_2)$. Then $0 = \alpha U(r_3) + (1 - \alpha)$, giving $U(r_3) = -\alpha/(1 - \alpha) < 0$.

Step 4. For $r_2 \prec r_3$: find α such that $r_2 \approx \alpha(r_1) + (1 - \alpha)(r_3)$. Then $1 = (1 - \alpha)U(r_3)$, giving $U(r_3) = 1/(1 - \alpha) > 1$.

Step 5 (Consistency). If $U(r_3), U(r_4), U(r_5)$ are found with $r_3 \prec r_4 \prec r_5$, find α with $r_4 \approx \alpha(r_3) + (1 - \alpha)(r_5)$. Check: $U(r_4) = \alpha U(r_3) + (1 - \alpha)U(r_5)$.

Example: football or movie? (Berger, Ex. 1) – setup

Saturday afternoon: a_1 = football game, a_2 = movie. There is a 40% chance of rain (θ_1).

Four possible outcomes (action, weather):

$r_1 = (a_1, \theta_1)$:	football, rain	(ruined afternoon – worst)
$r_2 = (a_1, \theta_2)$:	football, no rain	(great day – best)
$r_3 = (a_2, \theta_1)$:	movie, rain	(good movie, smart choice)
$r_4 = (a_2, \theta_2)$:	movie, no rain	(okay, slight regret)

Preference ordering: $r_1 \prec r_4 \prec r_3 \prec r_2$.

Fix the scale: $U(r_1) = 0$, $U(r_2) = 1$. We need to determine $U(r_3)$ and $U(r_4)$.

Determining the utilities: two subjective judgments

To find $U(r_4)$ and $U(r_3)$, the decision maker asks: "at what gamble between r_1 (worst) and r_2 (best) am I indifferent to this reward?"

Judgment 1: "I am indifferent between r_4 (movie, no rain) and a gamble that gives r_1 with probability 0.4 and r_2 with probability 0.6." That is: $r_4 \approx 0.4(r_1) + 0.6(r_2)$. This gives $U(r_4) = 0.4 \cdot 0 + 0.6 \cdot 1 = 0.6$.

Judgment 2: "I am indifferent between r_3 (movie, rain) and a gamble that gives r_1 with probability 0.3 and r_2 with probability 0.7." That is: $r_3 \approx 0.3(r_1) + 0.7(r_2)$. This gives $U(r_3) = 0.3 \cdot 0 + 0.7 \cdot 1 = 0.7$.

Both of these are subjective: they come from the decision maker's introspection. Different people could reasonably give different numbers.

The consistency check: a third judgment that is not free

We now have $U(r_1) = 0$, $U(r_4) = 0.6$, $U(r_3) = 0.7$, $U(r_2) = 1$.

Judgment 3: The decision maker compares r_3 to a gamble between r_4 and r_2. "At what α am I indifferent between r_3 and $\alpha(r_4) + (1-\alpha)(r_2)$?" After reflection: $\alpha = 0.6$.

But if the utility function from Judgments 1 and 2 is correct, then:

$$U(r_3) = \alpha\, U(r_4) + (1-\alpha)\, U(r_2) \quad \Longrightarrow \quad 0.7 = 0.6\alpha + (1-\alpha) \quad \Longrightarrow \quad \alpha = 0.75.$$

The decision maker said $\alpha = 0.6$, but the first two judgments force $\alpha = 0.75$. At least one of the three judgments is wrong.

This is the point of the consistency check: once two utilities are fixed by Judgments 1 and 2, the third comparison is *determined*. If it disagrees, the decision maker must go back and revise until all three are mutually consistent. Suppose the revised values are $U(r_4) = 0.6$ and $U(r_3) = 0.75$.

Making the decision

With the consistent utilities $U(r_1) = 0,\ U(r_3) = 0.75,\ U(r_4) = 0.6,\ U(r_2) = 1$:

Expected utility of each action:

$$\begin{aligned} a_1 \text{ (football)}: &\quad \pi(\theta_1)\, U(r_1) + \pi(\theta_2)\, U(r_2) = (0.4)(0) + (0.6)(1) = 0.6. \\ a_2 \text{ (movie)}: &\quad \pi(\theta_1)\, U(r_3) + \pi(\theta_2)\, U(r_4) = (0.4)(0.75) + (0.6)(0.6) = 0.66. \end{aligned}$$

The optimal action: go to the movie.

The raw payoffs might have suggested football (higher upside). But the utility function encodes an asymmetry: a ruined afternoon ($U = 0$) is much worse than the mild regret of missing the game when it did not rain ($U = 0.6$, not that bad). That asymmetry tips the balance.

The axioms

For the construction to work and for U to represent preferences, the preference ordering must be "rational." Four axioms:

Axiom 1 (Completeness): For any $P_1, P_2 \in \mathcal{P}$, either $P_1 \prec P_2$, $P_1 \approx P_2$, or $P_2 \prec P_1$.

Axiom 2 (Transitivity): If $P_1 \preceq P_2$ and $P_2 \preceq P_3$, then $P_1 \preceq P_3$.

Axiom 3 (Independence): If $P_1 \prec P_2$, then $\alpha P_1 + (1-\alpha)P_3 \prec \alpha P_2 + (1-\alpha)P_3$ for any $0 < \alpha < 1$ and $P_3 \in \mathcal{P}$.

Axiom 4 (Continuity): If $P_1 \prec P_2 \prec P_3$, there exist $0 < \alpha, \beta < 1$ such that $\alpha P_1 + (1-\alpha)P_3 \prec P_2$ and $P_2 \prec \beta P_1 + (1-\beta)P_3$.

Interpreting the axioms

Axioms 1 and 2 are housekeeping: you can compare anything, and your comparisons are transitive.

Axiom 3 (independence): if you prefer P_2 to P_1, mixing both with the same P_3 does not reverse the preference. Replacing P_1 by P_2 in one component of a mixture is always an improvement.

Axiom 4 (continuity): no consequence is infinitely good or infinitely bad. If P_1 were infinitely bad (say, death), there would be no $\beta > 0$ for which you would risk it to get P_3 instead of P_2. But if death were truly infinitely bad, you would never cross a street, never drive a car. People do these things, so Axiom 4 holds in practice.

The existence theorem and uniqueness

Theorem

If the preference ordering on $\mathcal{P}$ satisfies Axioms 1–4, there exists a utility function $U : \mathcal{R} \to \mathbb{R}$ such that $P_1 \prec P_2 \iff E^{P_1}[U(r)] < E^{P_2}[U(r)]$.

We will not prove existence. The proof constructs U using the procedure from earlier and verifies it represents the preferences; see DeGroot (1970). What we will prove are two properties of U that we will use.

Uniqueness up to affine transformation. If U is a utility function, so is $V = bU + c$ for any $b > 0$. *Proof:*

$$\begin{aligned} E^{P_1}[U(r)] < E^{P_2}[U(r)] &\iff b\,E^{P_1}[U(r)] + c < b\,E^{P_2}[U(r)] + c \\ &\iff E^{P_1}[V(r)] < E^{P_2}[V(r)]. \end{aligned}$$

The preferences are unchanged. $\square$

This means the choice of $U(r_1) = 0$ and $U(r_2) = 1$ in the construction is just a convention. Any two reference points with $U(r_1) < U(r_2)$ would give an equivalent utility function.

Axiom 4 implies boundedness

The existence theorem gives us a utility function U. We now show it must be bounded.

Proposition

If the preference ordering satisfies Axioms 1–4 and U is the resulting utility function, then U is bounded.

Proof. Suppose for contradiction that U is unbounded above: there exist rewards $s_1, s_2, \ldots \in \mathcal{R}$ with $U(s_n) \to \infty$.

Choose weights $p_n > 0$ with $\sum p_n = 1$ and $\sum p_n U(s_n) = \infty$. Define $Q = \sum p_n(s_n) \in \mathcal{P}$.

Now pick any two rewards $a \prec b$. Since $E^Q[U(r)] = \infty > U(b) > U(a)$, we have

$$(a) \prec (b) \prec Q.$$

Axiom 4 applied to this triple says: there exists $\alpha \in (0,1)$ such that $\alpha(a) + (1-\alpha)Q \prec (b)$, i.e., $E^{\alpha(a)+(1-\alpha)Q}[U(r)] < U(b)$.

But for any $\alpha \in (0,1)$:

$$E^{\alpha(a)+(1-\alpha)Q}[U(r)] = \alpha\, U(a) + (1-\alpha) \cdot \infty = \infty > U(b).$$

No such α exists. Contradiction. A symmetric argument handles U unbounded below.

Properties of monetary utility functions

When $\mathcal{R}$ is monetary, $U(r)$ typically has four features:

(i) Linear for small r. You would take a fair coin flip on ± 25 cents. So U is linear near $r = 0$. The "small" range depends on context: 25 cents for a student, millions for a large corporation.

(ii) Concave for $r > 0$. Marginal utility $U'(r)$ is decreasing, so $U''(r) < 0$. An additional \$100 matters more at \$0 than at \$1,000,000.

(iii) Different shape for gains and losses. People are often risk-seeking for losses (willing to gamble to avoid a sure loss) but risk-averse for gains. So $U(r)$ may be convex for $r < 0$ and concave for $r > 0$.

(iv) Bounded. Few would assign meaningful additional utility to money beyond $\$10^{100}$.

Defining the loss function

Each pair (θ, a) determines a consequence with utility $U(\theta, a)$. The loss function is

$$L(\theta, a) = -U(\theta, a).$$

Maximizing expected utility is the same as minimizing expected loss.

This is the answer to our second question: **why is expected loss the right criterion?** Because the utility axioms tell us that expected utility is the correct measure of value under uncertainty. Expected loss inherits this justification.

The Bayesian expected loss $\rho(\pi, a) = E_\pi[L(\theta, a)]$, the risk $R(\theta, \delta) = E_{P_\theta}[L(\theta, \delta(X))]$, the Bayes risk $r(\pi, \delta)$, and the loss of a randomized rule – all are justified by this construction.

When consequences are random

Often (θ, a) does not directly determine the consequence. The reward is a random variable Z whose distribution depends on θ and a. Then:

$$U(\theta, a) = E_{\theta,a}[U(Z)], \qquad L(\theta, a) = -E_{\theta,a}[U(Z)].$$

One must be careful to define Z as the *overall* outcome.

Properties of loss functions

1. Invariance under positive affine transformations.
Claim: If L is a valid loss, then $L' = bL + c$ (with $b > 0$) leads to the same optimal decisions.
Proof (for the Bayes case): The Bayes risk is $r(\pi, \delta) = E_\pi[R(\theta, \delta)]$. Under L':

$$r'(\pi, \delta) = E_\pi[E_{P_\theta}[bL(\theta, \delta(X)) + c]] = b\, r(\pi, \delta) + c.$$

Since $b > 0$, the δ minimizing r' is the same as the δ minimizing r. $\square$

So we can always shift: $L^*(\theta, a) = L(\theta, a) - \inf_{\theta,a} L(\theta, a) \geq 0$.

2. Regret loss. $\ell(\theta, a) = \sup_{a'} U(\theta, a') - U(\theta, a)$: the loss relative to the best action for the true θ. Equivalent to L for Bayesian analysis, but can differ for minimax. We return to this in Chapter 5.

I. Squared-error loss

$$L(\theta, a) = (\theta - a)^2.$$

Why is it popular?

- For unbiased estimators, $R(\theta, \delta) = \mathrm{Var}_{P_\theta}(\delta(X))$.
- Related to least squares theory.
- Calculations are often tractable.

Berger's honest assessment: "The above justifications really have very little merit. The question is – does squared-error loss typically reflect the true loss?"

Squared-error loss is unbounded and convex: large errors are penalized very severely. In many real problems, the loss should be bounded and (for large errors) concave.

Example: risk of the sample mean under squared-error loss

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\theta, \sigma^2)$. Estimate θ with $\delta(X) = \bar{X}$.

$$R(\theta, \delta) = E_{P_\theta}[(\theta - \bar{X})^2] = \mathrm{Var}_{P_\theta}(\bar{X}) = \frac{\sigma^2}{n}.$$

The risk is constant in θ and decreases in n.

Now consider $\delta_c(X) = c\bar{X}$ for some $c \in \mathbb{R}$. Then:

$$R(\theta, \delta_c) = E_{P_\theta}[(\theta - c\bar{X})^2] = c^2 \frac{\sigma^2}{n} + (1-c)^2\theta^2.$$

This is Example 4 from Lecture 1 with σ^2/n replacing 1. Minimizing over c for a fixed θ gives $c^* = \theta^2/(\theta^2 + \sigma^2/n)$, which depends on θ – so the "optimal shrinkage" is unknowable. This tension is at the heart of the course.

The Taylor expansion defense of squared-error loss

Suppose the reward is $Z = h(\theta - a, Y)$, where Y is an extraneous random variable (e.g., state of the economy) independent of θ and a.

Define $g(\theta - a, Y) = U(h(\theta - a, Y))$. If the sample information is accurate, $\theta - a$ is small. Expand in Taylor series around 0:

$$g(\theta - a, Y) \approx g(0, Y) + (\theta - a)g'(0, Y) + \tfrac{1}{2}(\theta - a)^2 g''(0, Y).$$

Taking expectations over Y, with $K_1 = -E_Y[g(0, Y)]$, $K_2 = -E_Y[g'(0, Y)]$, $K_3 = -\frac{1}{2}E_Y[g''(0, Y)]$:

$$L(\theta, a) = -E_Y[g(\theta - a, Y)] \approx K_1 + K_2(\theta - a) + K_3(\theta - a)^2.$$

Completing the square (assuming $K_3 > 0$):

$$L(\theta, a) \approx K_3\left(\theta - a + \frac{K_2}{2K_3}\right)^2 + \text{const.}$$

When $K_2 = 0$ (symmetric utility in y, symmetric Y), this is squared-error loss. When $K_2 \neq 0$, it is shifted squared-error: $(\theta - a + c)^2$, which reduces to standard squared-error by redefining $\mathcal{A}$.

Variants of squared-error loss

Weighted squared-error: $L(\theta, a) = w(\theta)(\theta - a)^2$.
The weight $w(\theta) > 0$ encodes that a given error may matter more at some parameter values than others.

Example: estimating a Poisson rate $\theta > 0$. An error of 1 when $\theta = 0.5$ is much worse than the same error when $\theta = 100$. The weight $w(\theta) = 1/\theta^2$ normalizes for scale, giving $L(\theta, a) = (1 - a/\theta)^2$.

Quadratic loss (multivariate): $\theta, a \in \mathbb{R}^p$, Q positive definite:

$$L(\theta, a) = (\theta - a)^T Q(\theta - a).$$

When $Q = I_p$: $L(\theta, a) = \|\theta - a\|^2 = \sum_{i=1}^{p} (\theta_i - a_i)^2$.

We will encounter this extensively when studying Stein estimation in Week 6.

Example: risk under sum-of-squares loss

$X = (X_1, \ldots, X_p)^T$ with $X_i \overset{\text{ind}}{\sim} \mathcal{N}(\theta_i, 1)$. Estimate $\theta = (\theta_1, \ldots, \theta_p)^T$ under $L(\theta, a) = \|\theta - a\|^2$.

The natural estimator is $\delta(X) = X$. Its risk:

$$R(\theta, \delta) = E_{P_\theta}\left[\sum_{i=1}^{p}(\theta_i - X_i)^2\right] = \sum_{i=1}^{p} \mathrm{Var}_{P_\theta}(X_i) = p.$$

Constant risk p, for all θ. Seems hard to improve.

For $p \geq 3$, James and Stein (1960) showed the estimator

$$\delta^{JS}(X) = \left(1 - \frac{p-2}{\|X\|^2}\right) X$$

satisfies $R(\theta, \delta^{JS}) < p$ for *all* θ. The sample mean vector is inadmissible.

That result is one of the major topics later in the course. For now, it illustrates that the choice of loss (here, sum-of-squares) can have dramatic consequences for which estimators are admissible.

II. Linear loss

When the utility function is approximately linear over the relevant range, the loss is approximately linear:

$$L(\theta, a) = \begin{cases} K_0(\theta - a) & \text{if } \theta - a \geq 0 \quad \text{(underestimation)} \\ K_1(a - \theta) & \text{if } \theta - a < 0 \quad \text{(overestimation)} \end{cases}$$

K_0 and K_1 encode the relative cost of under- vs. overestimation.
When $K_0 = K_1$: **absolute error loss** $L(\theta, a) = |\theta - a|$.

Recall Lecture 1, Example 1 (drug company): $L(\theta_2, a) = (\theta_2 - a)$ if $\theta_2 \geq a$ and $2(a - \theta_2)$ if $\theta_2 < a$. That is linear loss with $K_0 = 1$, $K_1 = 2$: overestimating the adverse reaction rate costs twice as much as underestimating.

Example: car production under linear loss (Berger, Ex. 17)

A company must decide how many cars a to produce. The demand is θ (unknown). Each car sold yields \$500 profit. Each unsold car yields \$1000 loss.

If $a \leq \theta$: all cars sell, profit $= 500a$. If $a > \theta$: θ cars sell, $a - \theta$ do not, profit $= 500\theta - 1000(a - \theta) = 1500\theta - 1000a$.

The regret loss (loss relative to the best action if you knew θ): the best action is $a^* = \theta$ (produce exactly the demand, profit 500θ). So:

$$\ell(\theta, a) = 500\theta - \text{profit}(a) = \begin{cases} 500(\theta - a) & \text{if } a \leq \theta \\ 1500(a - \theta) & \text{if } a > \theta \end{cases}$$

This is linear loss with $K_0 = 500$ (opportunity cost of underproduction) and $K_1 = 1500$ (cost of overproduction). The asymmetry $K_1/K_0 = 3$ reflects that unsold cars are much worse than unmet demand.

III. "0–1" loss

In a two-action problem (testing $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$):

$$L(\theta, a_i) = \begin{cases} 0 & \text{if } \theta \in \Theta_i \quad \text{(correct)} \\ 1 & \text{if } \theta \in \Theta_j,\ j \neq i \quad \text{(incorrect)} \end{cases}$$

The risk function:

- When $\theta \in \Theta_0$: $R(\theta, \delta) = P_{P_\theta}(\text{reject } H_0) = \text{Type I error}$.
- When $\theta \in \Theta_1$: $R(\theta, \delta) = P_{P_\theta}(\text{accept } H_0) = \text{Type II error}$.

The Bayesian expected loss: $\rho(\pi^*, a_i) = 1 - \pi^*(\theta \in \Theta_i)$.

The error probabilities from Neyman–Pearson theory are the risk function under 0–1 loss. Testing is a special case of decision theory.

Example: 0–1 loss for a normal test

$X \sim \mathcal{N}(\theta, 1)$. Test $H_0 : \theta \leq 0$ vs. $H_1 : \theta > 0$. Decision rule: reject H_0 if $X > c$.

Under 0–1 loss, the risk function is:

$$R(\theta, \delta_c) = \begin{cases} P_{P_\theta}(X > c) = 1 - \Phi(c - \theta) & \text{if } \theta \leq 0 \quad \text{(Type I)} \\ P_{P_\theta}(X \leq c) = \Phi(c - \theta) & \text{if } \theta > 0 \quad \text{(Type II)} \end{cases}$$

For $c = 0$: Type I error at $\theta = 0$ is $1 - \Phi(0) = 0.5$. Type II error at $\theta = 1$ is $\Phi(-1) = 0.159$. For $c = 1.645$: Type I error at $\theta = 0$ is $1 - \Phi(1.645) = 0.05$. Type II error at $\theta = 1$ is $\Phi(0.645) = 0.74$.

The familiar tradeoff between Type I and Type II error is the tradeoff between the risk function at different values of θ: making $R(\theta, \delta_c)$ smaller for $\theta \leq 0$ makes it larger for $\theta > 0$, and vice versa.

Weighted 0–1 loss

Basic 0–1 loss treats the two types of error as equally costly. Rarely appropriate.

More realistic:

$$L(\theta, a_i) = \begin{cases} 0 & \text{if correct} \\ K_0 & \text{if reject } H_0 \text{ when } \theta \in \Theta_0 \quad \text{(Type I)} \\ K_1 & \text{if accept } H_0 \text{ when } \theta \in \Theta_1 \quad \text{(Type II)} \end{cases}$$

The risk becomes $K_0 \cdot P_{P_\theta}(\text{reject } H_0)$ for $\theta \in \Theta_0$ and $K_1 \cdot P_{P_\theta}(\text{accept } H_0)$ for $\theta \in \Theta_1$.

Even more generally: $L(\theta, a_1) = K_0(\theta)$ for $\theta \in \Theta_0$, where the weight varies within Θ_0. A Type I error may be worse when θ is far from the boundary.

Inference as a decision problem

The most common use: representing standard inference measures as a risk or Bayesian expected loss.

Confidence sets. Let $C(x) \subseteq \Theta$ be a confidence rule. The loss

$$L(\theta, C(x)) = 1 - I_{C(x)}(\theta) = \begin{cases} 0 & \text{if } \theta \in C(x) \\ 1 & \text{if } \theta \notin C(x) \end{cases}$$

gives $R(\theta, C) = 1 - P_{P_\theta}(C(X) \ni \theta)$: one minus the coverage probability.

Unbiased estimation. Restrict to $E_{P_\theta}[\delta(X)] = \theta$ for all θ. Under squared-error loss, $R(\theta, \delta) = \mathrm{Var}_{P_\theta}(\delta(X))$. The UMVUE is the decision-theoretic optimum within this restricted class.

Scoring rules: quantifying how well confidence works (Berger, Ex. 4)

A confidence rule $C(x)$ with associated "confidence" $\alpha(x)$: after observing x, we report $\alpha(x)$ as our confidence that $\theta \in C(x)$. How well does $\alpha(x)$ reflect reality?

The Brier scoring rule:

$$L_C(\theta, \alpha(x)) = (\alpha(x) - I_{C(x)}(\theta))^2.$$

When $\theta \in C(x)$: loss $= (\alpha(x) - 1)^2$, small if $\alpha(x) \approx 1$.
When $\theta \notin C(x)$: loss $= \alpha(x)^2$, small if $\alpha(x) \approx 0$.

Scoring rules continued: constant vs. data-dependent confidence

Consider two choices: $\alpha_1(x) \equiv 0.75$ (constant frequentist confidence) and $\alpha_2(x) = P(\theta \in C(x) \mid x)$ (posterior probability, a data-dependent confidence).

The risk of α_1 under the Brier loss:

$$R(\theta, \alpha_1) = E_{P_\theta}\left[(\alpha_1(X) - I_{C(X)}(\theta))^2\right] = 0.75^2 \cdot P_{P_\theta}(\theta \notin C(X)) + 0.25^2 \cdot P_{P_\theta}(\theta \in C(X)).$$

If C has 75% coverage: $R(\theta, \alpha_1) = (0.5625)(0.25) + (0.0625)(0.75) = 0.1875$.

The data-dependent confidence α_2 adapts: it reports high confidence when the data are informative and low confidence when they are not. One can show (and we will, in Chapter 4) that α_2 has $R(\theta, \alpha_2) \leq R(\theta, \alpha_1)$ for all θ, with strict inequality for some θ.

The data-dependent confidence *dominates* the constant one: it communicates better what $C(x)$ is worth for the specific x you observed.

Predictive problems

Predict a future observation Z with density $g(z|\theta)$, given data X with density $f(x|\theta)$.

If the original loss is $L^*(z, a)$, the decision-theoretic reduction is:

$$L(\theta, a) = E_{P_\theta}[L^*(Z, a)] = \int L^*(z, a)\, g(z|\theta)\, dz.$$

This reduces the prediction problem to a standard decision problem involving θ.

Example: predicting a normal (Berger, Ex. 5)

$Z \sim \mathcal{N}(\theta, \sigma^2)$, $L^*(z, a) = (z - a)^2$. Then:

$$\begin{aligned} L(\theta, a) &= E_{P_\theta}[(Z - a)^2] = E_{P_\theta}[(Z - \theta)^2 + 2(Z - \theta)(\theta - a) + (\theta - a)^2] \\ &= \mathrm{Var}_{P_\theta}(Z) + 0 + (\theta - a)^2 = \sigma^2 + (\theta - a)^2. \end{aligned}$$

The cross term vanishes because $E_{P_\theta}[Z - \theta] = 0$.

The constant σ^2 does not affect the choice of a. Choosing the best prediction of Z is equivalent to estimating θ under squared-error loss.

This partially explains why decision theorists focus on estimating unknown parameters rather than predicting future observations: prediction problems often reduce to parameter estimation through the loss.